

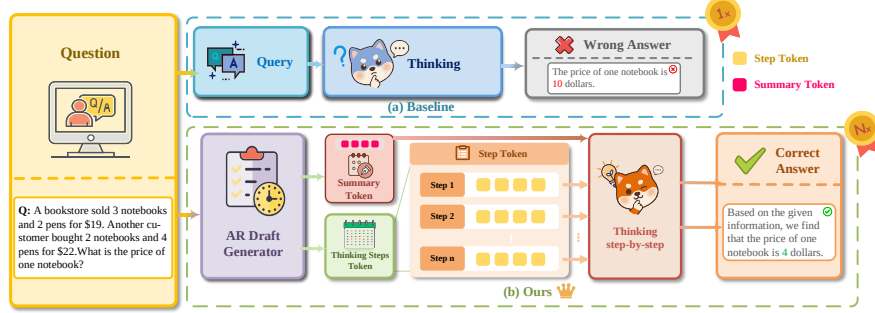


Figure 1: Comparative visualization of the baseline and our Cross-dLLM reasoning pipelines. (a) The baseline approach performs sequential thinking, which leads to a wrong answer. (b) Our Cross-dLLM method employs an AR Draft Generator with parallel step tokens and summary/thinking tokens, enabling step-by-step reasoning that produces the correct answer.

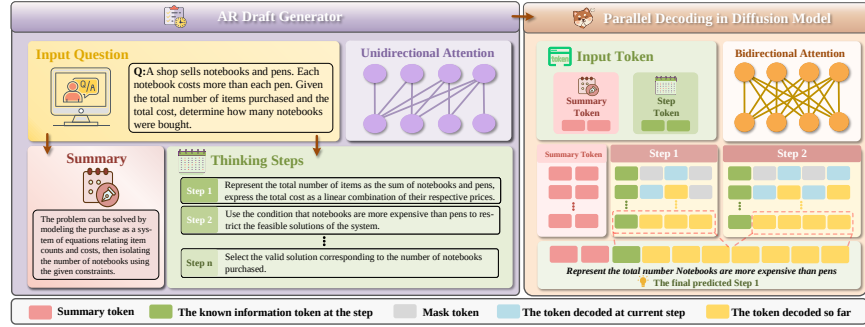


Figure 2: Overview of the Cross-dLLM method: Autoregressive drafting for accelerated diffusion reasoning. This figure illustrates the internal mechanism of our proposed Cross-dLLM, where an autoregressive draft generator first creates a structured semantic skeleton, which is then used by the diffusion model for parallel decoding of detailed reasoning steps. (Left) The AR Draft Generator takes the input question and employs unidirectional attention to produce a Summary token followed by a sequence of Thinking Steps (Step 1 to Step n). (Right) The diffusion model performs parallel decoding using bidirectional attention on the Summary Token and Step Tokens, with color-coded token types: pink for Summary token, green for known information at the step, gray for mask token, blue for the token decoded at the current step, and yellow for tokens decoded so far.

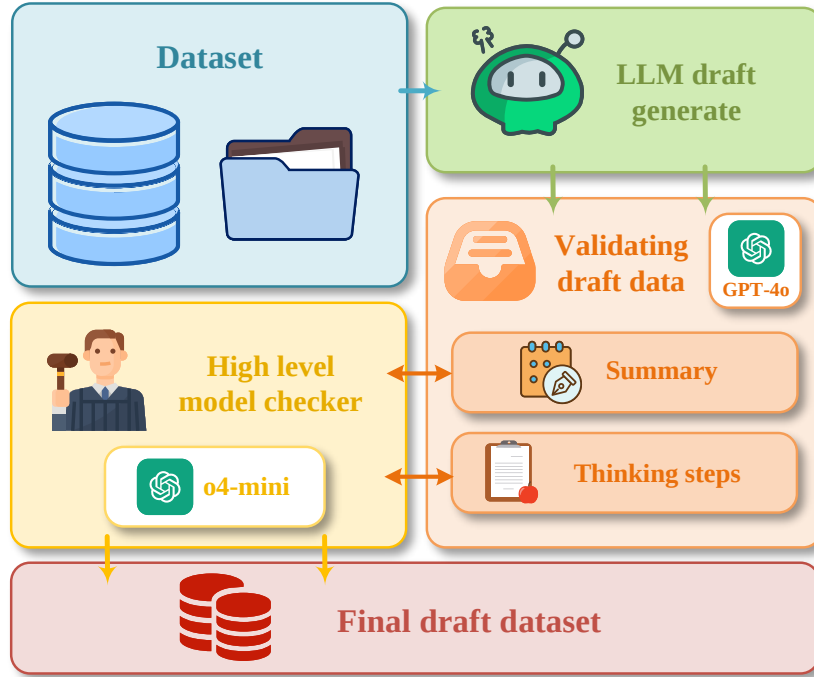


Figure 3: Dataset construction pipeline for the proposed Cross-dLLM. This figure illustrates the overall process of building the high-quality draft dataset. Starting from the original Dataset, an LLM draft generator creates initial reasoning drafts. The drafts undergo validation using GPT-4o to produce Summary and Thinking steps. A high-level model checker (powered by o4-mini) then performs additional validation and refinement. The final outputs are collected into the Final draft dataset.